A. World Football Cup

time limit per test: 2 seconds
memory limit per test: 64 megabytes

Everyone knows that 2010 FIFA World Cup is being held in South Africa now. By the decision of BFA (Berland's Football Association) next World Cup will be held in Berland. BFA took the decision to change some World Cup regulations:

- the final tournament features n teams (n is always even)
- the first $n / 2$ teams (according to the standings) come through to the knockout stage
- the standings are made on the following principle: for a victory a team gets 3 points, for a draw — 1 point, for a defeat — 0 points. In the first place, teams are ordered in the standings in decreasing order of their points; in the second place — in decreasing order of the difference between scored and missed goals; in the third place — in the decreasing order of scored goals
- it's written in Berland's Constitution that the previous regulation helps to order the teams without ambiguity.

You are asked to write a program that, by the given list of the competing teams and the results of all the matches, will find the list of teams that managed to get through to the knockout stage.

Input

The first input line contains the only integer n $(1 \le n \le 50)$ — amount of the teams, taking part in the final tournament of World Cup. The following n lines contain the names of these teams, a name is a string of lower-case and upper-case Latin letters, its length doesn't exceed 30 characters. The following $n \cdot (n - 1) / 2$ lines describe the held matches in the format `name1-name2 num1:num2`, where $name1$, $name2$ — names of the teams; $num1$, $num2$ $(0 \le num1, num2 \le 100)$ — amount of the goals, scored by the corresponding teams. Accuracy of the descriptions is guaranteed: there are no two team names coinciding accurate to the letters' case; there is no match, where a team plays with itself; each match is met in the descriptions only once.

Output

Output $n / 2$ lines — names of the teams, which managed to get through to the knockout stage in lexicographical order. Output each name in a separate line. No odd characters (including spaces) are allowed. It's guaranteed that the described regulations help to order the teams without ambiguity.

Examples

input Copy

```
4
A
B
C
D
A-B 1:1
A-C 2:2
A-D 1:0
B-C 1:0
B-D 0:3
C-D 0:3
```

output Copy

```
A
D
```

input Copy

```
2
a
```

→ **Attention**

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 19

Finished

Practice

→ **Virtual participation**

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
371197025	Apr/15/2026 06:08	Accepted

→ **Problem tags**

implementation *1400

No tag edit access

```
A
a-A 2:1
```

output Copy

```
a
```

→ Contest materials

- Announcement
- Tutorial (en)

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The only programming contests Web 2.0 platform
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